Gio Vase Dashboard

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CHAPTER

ONE

GIOVASE DASHBOARD - DATABASE

class model.gio_db.**ConnectionRequest**(***kwargs*)
A class used to represent a connection request from plant

Params:

- id : String, vase serial number
- signal: int, the value of the request's signal, ranges from -128 to -42. -128 means a weak signal and -42 means a strong one.
- pairing: int, three-digit random integer, representing the connection request of the plant.
- radio_id : String, radio serial number

class model.gio_db.**Plant**(***kwargs*)
A class used to represent the plant

class model.gio_db.**Radio**(***kwargs*)
A class used to represent the radio-raspberry

Params:

- id: String, radio serial number
- name: string, radio name
- url_radio: string, raspberry url used to communicate operation request to the radio and to the associated plants
- transmit_power : int, radio transmission power, range 0-7 (lower, higher)
- sleep_time: time interval in which the radio is in sleep state (minutes)

class model.gio_db.**TypePlant**(***kwargs*)
Rappresents the type of plant

class model.gio_db.**User**(**args*, ***kw*)
Rappresents one user.

Params:

- id
- username
- password
- role
- is_active: boolean
- is_anonymous: boolean

authenticate(*password*)
if the password provided is correct, it authenticates the user :param: password: user password :type: password: string :return: True or False

get_id()

Returns the user id

Return type Integer

property is_admin
check if the user is admin :return: True or False

is_authenticated()
it returns true if the user is already authenticated, otherwise false :return: self.is_authenticated :rtype: true or false

set_password(*password*)
set the password of the user :param: password: user password :type: password: string

set_role(*role*)
It sets the user role :param role: the role :type role: value of the enum Role :return: True if the user's role has been set, otherwise false :rtype: Boolean

set_username(*username*)
it sets the user username :param username: new username :type username: string :return: true if username has been set, otherwise false :rtype: Boolean

model.gio_db.**add_conn_req**(*idv*, *signalv*, *pairingv*, *radio_id*)

- **if there is no connection request or a plant with id equal to the idv parameter:** it adds the connection request and returns the value True.
- **If the connection request with idv was already present:** controls the signal strength of both requests (new and old). If the signal strength of the new request is greater, it replaces the old one with the new one and sends the refusal operation to the radio of the old connection request, otherwise only the pairing number changes. In both cases the function returns the value True.
- **If there was already a plant with id equal to idv (parameter):** send the join request to the associated radio: if this request is not successful, the plant is deleted and the new connection request is inserted, otherwise it does nothing. In this case the function returns the value False.

Parameters

- **idv** (*string*) – vase serial number
- **signalv** (*int*) – the value of the request's signal.
- **pairingv** (*int*) – three-digit random integer
- **radio_id** (*string*) – radio serial number

Returns True - if there was no connection request with id equal to idv, or if it was present, but it has been replaced because the signal of the new request is stronger. False - if a plant with id equal to idv was present and cannot communicate with the associated radio.

Return type boolean

model.gio_db.**add_plant**(*idv*, *radio*)
if the radio with serial number equal to the second parameter exists and the plant doesn't exist, adds the plant with serial number equal to idv, and deletes the connection request with id equal to idv.

Parameters

- **idv** (*string*) – plant serial number
- **radio** (*string*) – radio serial number

model.gio_db.**add_radio**(*radio*, *url_radio*)

- **If the radio is not in the database:** add it, and return the value True.
- **If the radio is already present:**
  - if the url has not changed: it does nothing and returns the False value,
  - if the url has changed: change url and reset all the parameters, return the False value

Parameters

- **radio** (*str*) – serial number
- **url_radio** (*str*) – raspberry (radio) url

Returns True or False

Return type bool

model.gio_db.**add_type**(*idt*, *hum_min*, *hum_max*, *temp_min*, *temp_max*, *light_max*, *light_min*)
Adds a new type of plant

Parameters

- **idt** (*str*) – id (name) of the type
- **hum_min** (*int*) – min. humidity of the type
- **hum_max** (*int*) – max. humidity of the type
- **temp_min** (*int*) – min. temperature of the type
- **temp_max** (*int*) – max. temperature of the type
- **light_max** (*int*) – max. light of the type
- **light_min** (*int*) – min. light of the type

model.gio_db.**add_user**(*username*, *password*, *role*)
Add a user into the database

:param username :param password :param role :return True if the operation was successful otherwise False

model.gio_db.**associated_plants**(*radio_id*)
Gets plants list associated with the radio

Parameters **radio_id** – radio serial number

Returns the list of plants associated with the radio

Return type plant list

model.gio_db.**change_type**(*idv*, *idt*, *url*)
Change the type of the plant

Parameters

- **idv** (*str*) – plant serial number
- **idt** (*int*) – type of plant id
- **url** (*str*) – associated radio's url

Returns True in case of success, otherwise False

Return type bool

model.gio_db.**correct_password_user**(*username*, *password*)

Parameters

- **username** (*string*) –
- **password** (*string*) –

Returns True if the user's password is correct otherwise False

Return type boolean

model.gio_db.**delete_conn_req**(*idv*, *radio*)

if present, it eliminates the connection request with id equal to the first parameter from the radio with id equal to the second parameter.

Parameters

- **idv** (*string*) – connection (plant) serial number
- **radio** (*string*) – radio serial number

Returns None or the deleted request

model.gio_db.**delete_plant**(*idv*)

Delete the plant with serial number equal to idv, if presents.

Parameters **idv** (*str*) – plant serial number

model.gio_db.**delete_radio**(*radio*)

Delete the radio with serial number equal to parameter

Parameters **radio** (*str*) – serial number

Returns True or False

Return type bool

model.gio_db.**delete_user**(*username*, *password*)

Delete the user username.

Parameters

- **username** (*str*) –
- **password** (*str*) –

Returns True if the operation was successful otherwise False

Return type bool

model.gio_db.**get_user**(*username*)

Parameters **username** –

:return the user with username equal to the parameter, if it exist, otherwise none.

model.gio_db.**radio_from_url**(*url*)

Gets the radio with url equal to parameter.

Parameters **url** (*str*) – raspberry-radio url

Returns object radio or None

Return type radio

model.gio_db.**update_name**(*idv*, *name*)
Change the plant's name.

Parameters

- **idv** (*str*) – plant serial number
- **name** (*str*) – new name

Returns True (success) or False

Return type bool

model.gio_db.**update_plant_state_fitness**(*idv*)

Update the plant status, calculated with the values of:

- ideal humidity, ideal light, ideal temperature
- current humidity, current light, current temperature

Parameters **idv** (*str*) – serial number

Returns the fitness state

Return type float

model.gio_db.**update_radio_name**(*radio*, *name*)
Change the name of the radio

Parameters

- **radio** (*string*) – radio serial number
- **name** (*string*) – name to associate with the radio

Returns True if the operation is successful, otherwise False

Return type boolean

model.gio_db.**update_radio_transmit_power**(*radio*, *tr*)
Change the transmit-power of the radio

Parameters

- **radio** (*str*) – radio serial number
- **tr** (*int*) – new transmit power

Returns True or False

Return type bool

model.gio_db.**update_sleep_time**(*radio*, *time*)
Change the sleep-time of the radio

Parameters

- **radio** (*str*) – radio serial number
- **time** (*int*) – new sleep time (minutes)

Returns True or False

Return type bool

model.gio_db.**update_vase_transmit_power**(*idv*, *tr*)
Change the radio transmit power of the plant

Parameters

- **idv** (*str*) – serial number
- **tr** (*int*) – value of the new transmit power

Returns True or False

Return type bool

model.gio_db.**url_from_plant**(*idv*)

Gets the url of the associated raspberry-radio.

Parameters **idv** (*str*) – plant serial number

Returns url or -1

Return type str

model.gio_db.**url_from_radio**(*radio*)

Gets the radio url.

Parameters **radio** (*string*) – serial number

Returns the url of the radio or None

Return type string

CHAPTER

TWO

GIOVASE DASHBOARD - REQUEST

controller.request.**request**()

Recive a request from a Radio-Raspberry

- **Types of request:**
  - Operation.RADIO_JOIN

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